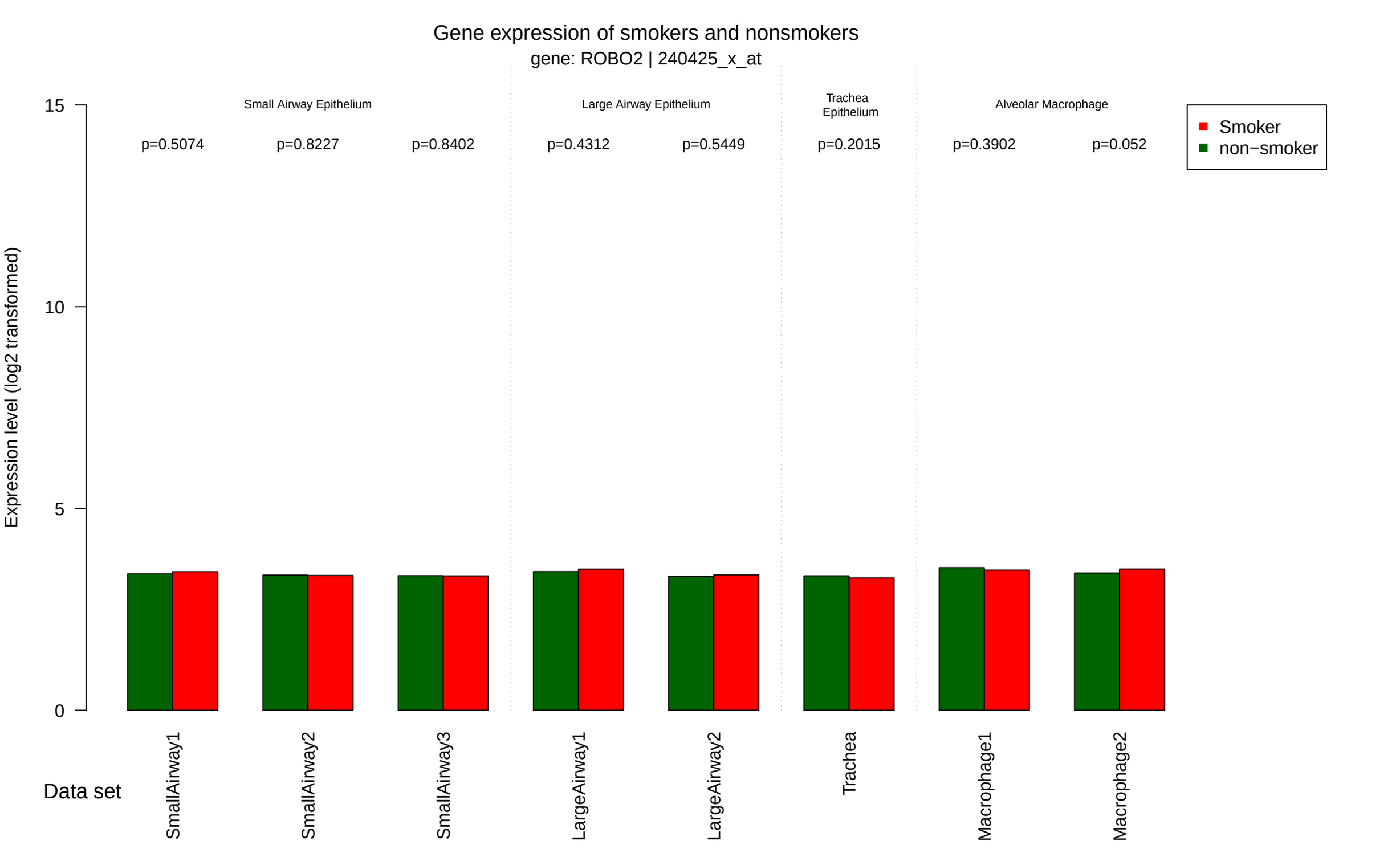
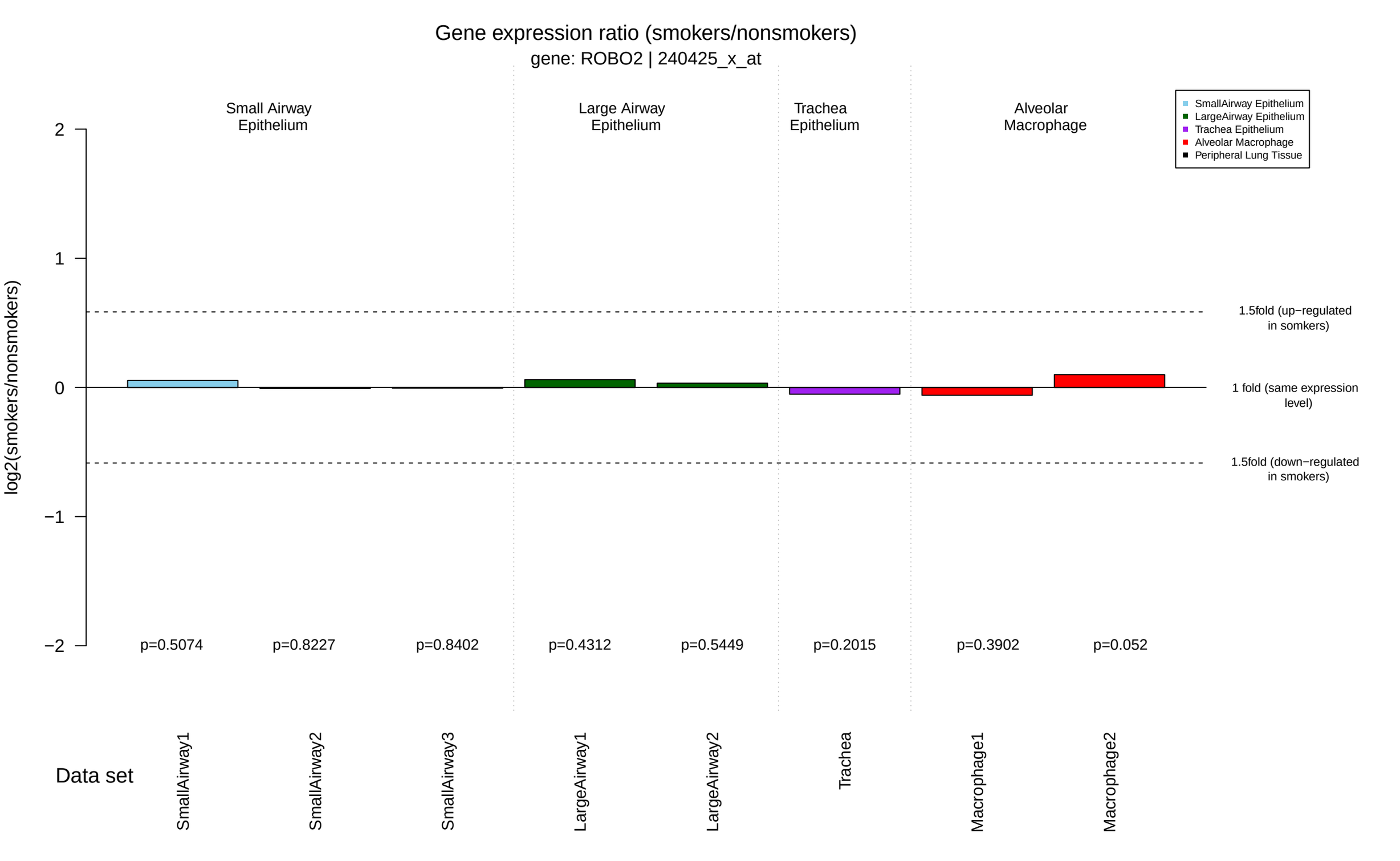
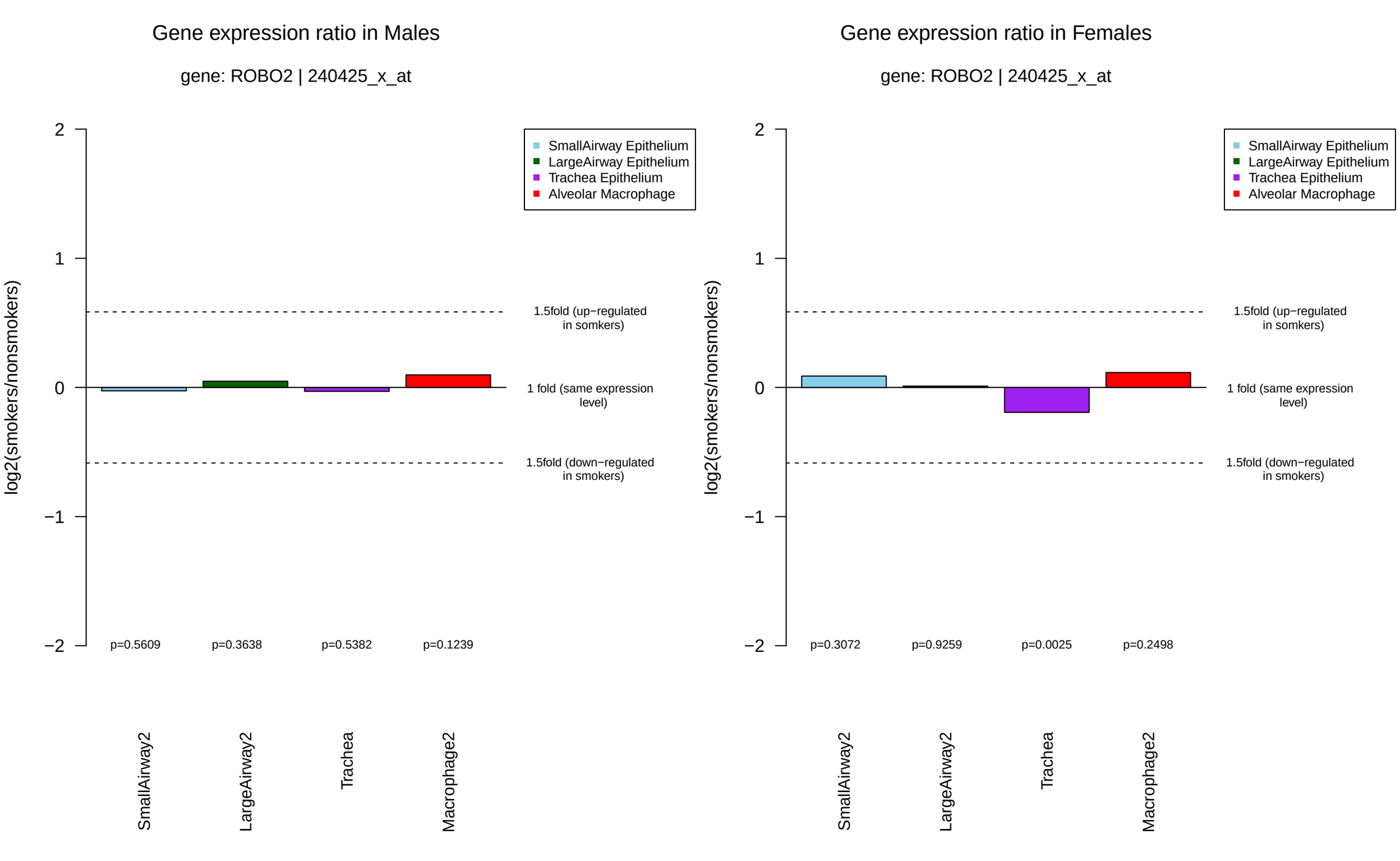
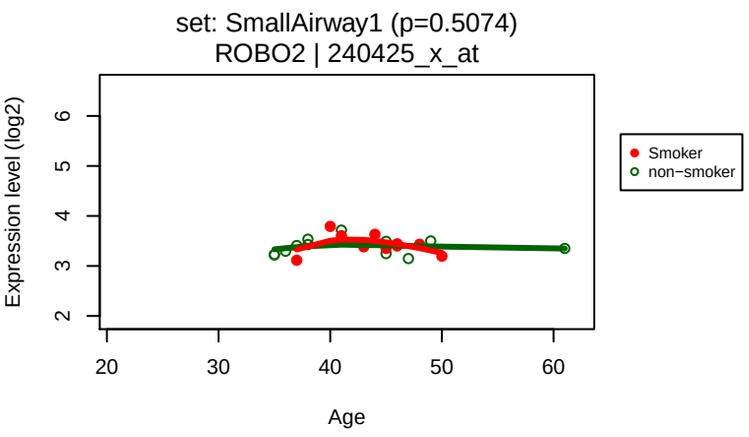
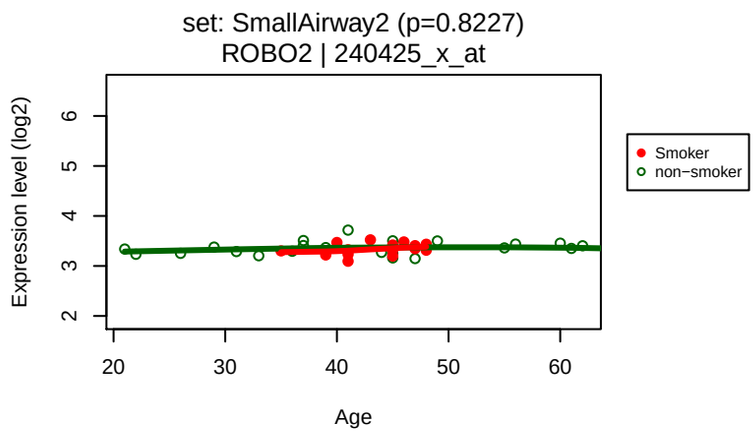
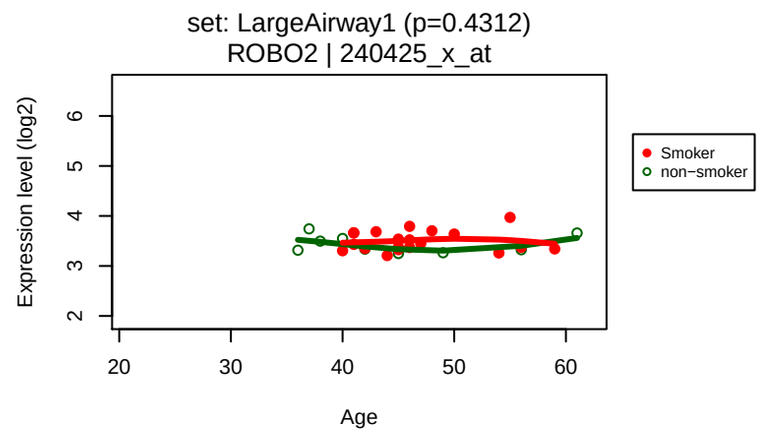

Gene expression of smokers and nonsmokers
gene: ROBO2 | 240425_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5074
p=0.8227
p=0.8402
p=0.4312
p=0.5449
p=0.2015
p=0.3902
p=0.052
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

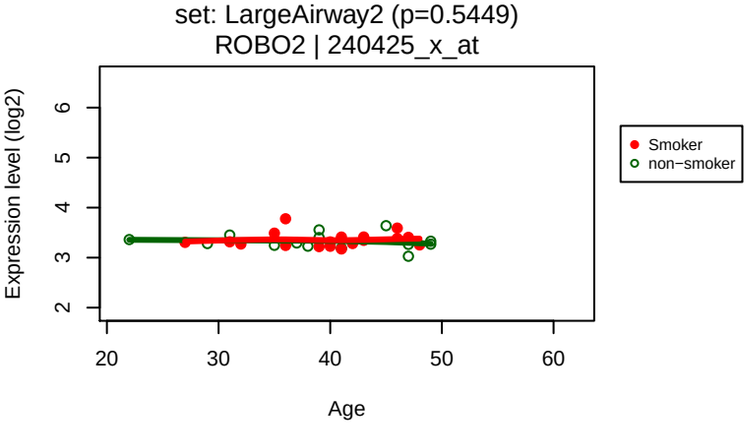

Gene expression ratio (smokers/nonsmokers)
gene: ROBO2 | 240425_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5074
p=0.8227
p=0.8402
p=0.4312
p=0.5449
p=0.2015
p=0.3902
p=0.052
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

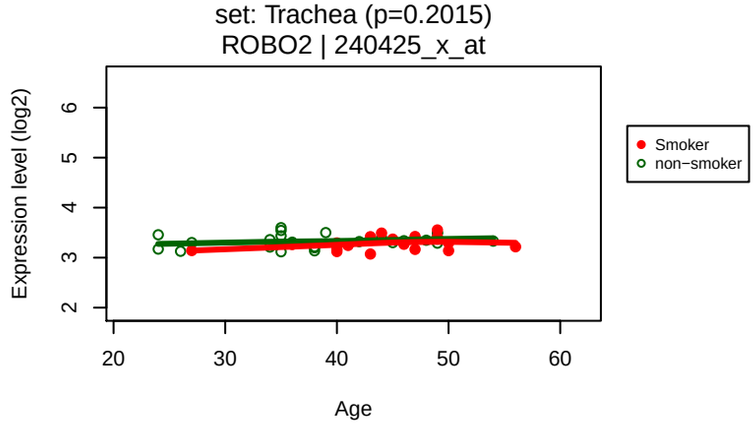

Gene expression ratio in Males
gene: ROBO2 | 240425_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5609
p=0.3638
p=0.5382
p=0.1239
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ROBO2 | 240425_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3072
p=0.9259
p=0.0025
p=0.2498
SmallAirway2
LargeAirway2
Trachea
Macrophage2

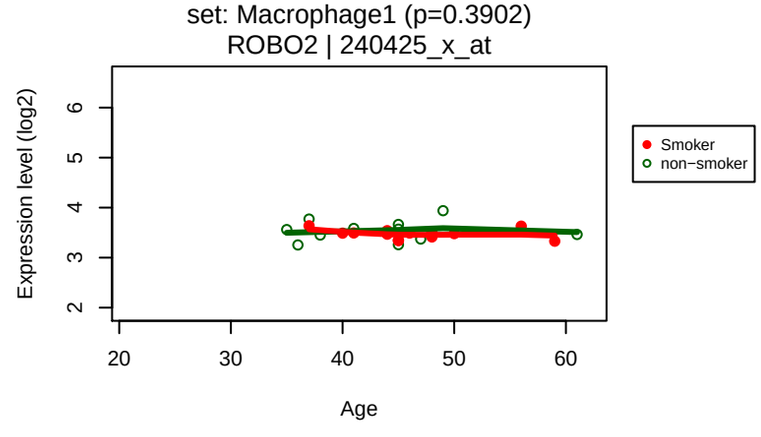

set: SmallAirway1 (p=0.5074)
ROBO2 | 240425_x_at
Expression level (log2)
Age
Smoker
non−smoker

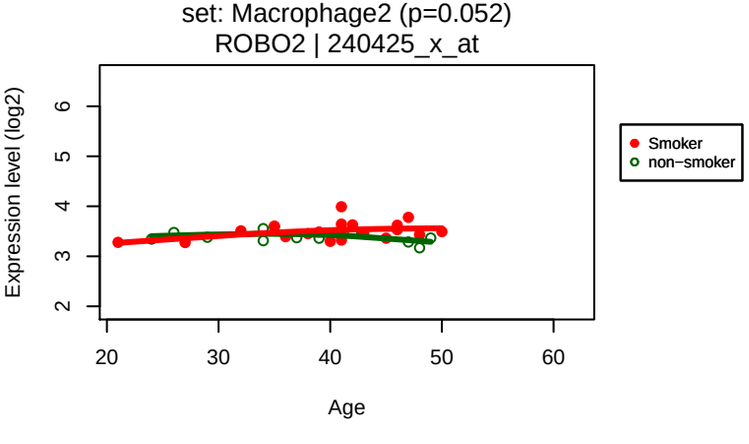

set: SmallAirway2 (p=0.8227)
ROBO2 | 240425_x_at
Expression level (log2)
Age
Smoker
non−smoker

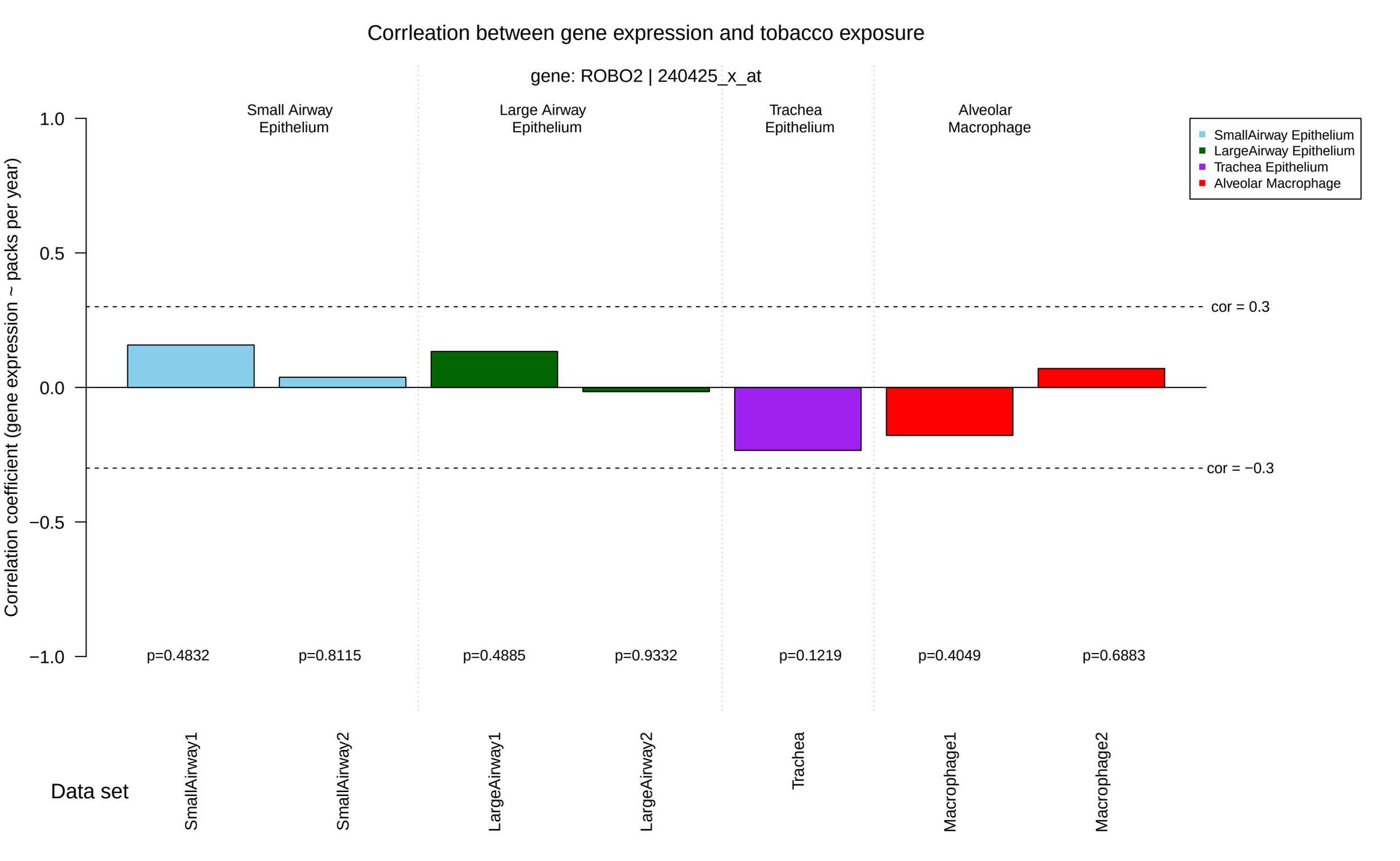

set: LargeAirway1 (p=0.4312)
ROBO2 | 240425_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5449)
ROBO2 | 240425_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2015)
ROBO2 | 240425_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3902)
ROBO2 | 240425_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.052)
ROBO2 | 240425_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ROBO2 | 240425_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4832
p=0.8115
p=0.4885
p=0.9332
p=0.1219
p=0.4049
p=0.6883
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2